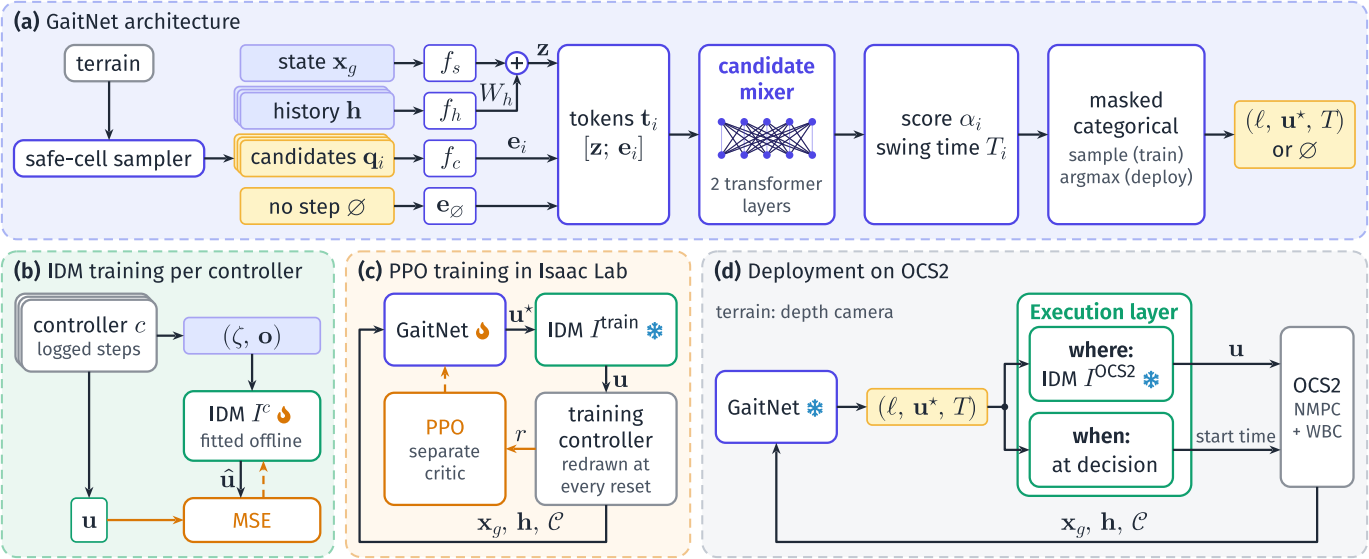

(a) GaitNet architecture
terrain
safe-cell sampler
state $\mathbf{x}_g$
history $\mathbf{h}$
candidates $\mathbf{q}_i$
no step $\varnothing$
f_s
f_h
f_c
$\mathbf{e}_\varnothing$
W_h
$\mathbf{e}_i$
$\mathbf{z}$
tokens $\mathbf{t}_i$ $[\mathbf{z};\ \mathbf{e}_i]$
candidate mixer
2 transformer layers
score α_i swing time T_i
masked categorical
sample (train)
argmax (deploy)
$(\ell,\ \mathbf{u}^\star,\ T)$ or $\varnothing$
(b) IDM training per controller
controller c logged steps
$(\zeta,\ \mathbf{o})$
IDM I^c fitted offline
$\hat{\mathbf{u}}$
$\mathbf{u}$
MSE
(c) PPO training in Isaac Lab
GaitNet
$\mathbf{u}^\star$
IDM I^{train}
$\mathbf{u}$
PPO separate critic
r
training controller redrawn at every reset
$\mathbf{x}_g,\ \mathbf{h},\ \mathcal{C}$
(d) Deployment on OCS2
terrain: depth camera
GaitNet
$(\ell,\ \mathbf{u}^\star,\ T)$
Execution layer
where: IDM I^{OCS2}
when: at decision
$\mathbf{u}$
start time
OCS2 NMPC + WBC
$\mathbf{x}_g,\ \mathbf{h},\ \mathcal{C}$